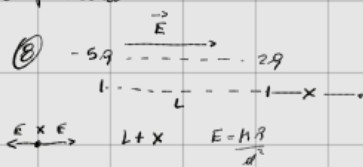


* Campo elétrico:



$$\frac{k \cdot 5q}{(L+x)^2} = \frac{k \cdot 2q}{(x)^2}$$

$$\frac{5}{(L+x)^2} = \frac{2}{x^2}$$

$$\rightarrow x \cdot \sqrt{5} = \sqrt{2} \cdot (L+x)$$

$$x \sqrt{5} = \sqrt{2} L + \sqrt{2} x$$

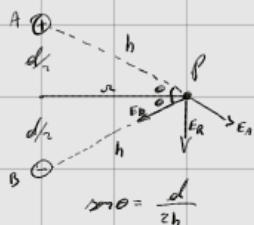
$$x (\sqrt{5} - \sqrt{2}) = \sqrt{2} L \rightarrow$$

$$x = \frac{\sqrt{2} L}{\sqrt{5} - \sqrt{2}}$$

10 $E_R = E_3 - E_1$

$$F_R = \frac{k \cdot (5e)}{5 \cdot 10^{-6}} - \frac{k \cdot (2e)}{10 \cdot 10^{-6}} = k \cdot e \cdot \left(\frac{1}{10^{-6}} - \frac{2}{5 \cdot 10^{-6}} \right) = -2,88 \cdot 10^{-1}$$

19



$$h = \sqrt{\frac{d^2}{4} + x^2}$$

$$b) \vec{E} = \frac{2kqd}{2x^2}$$

$$E_R = \frac{kqd}{\left(\frac{d^2}{4} + x^2\right)^{3/2}}$$

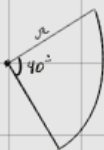
$$\vec{E} = 2 \cdot \left(\frac{kq}{h^2} \cdot \frac{d}{2h} \right) = \frac{2kqd}{\left(\frac{d^2}{4} + x^2\right)^{3/2}} = \frac{2kqd}{2\left(\frac{d^2}{4} + x^2\right)^{3/2}} = \frac{kqd}{\left(\frac{d^2}{4} + x^2\right)^{3/2}}$$

$$* \left(\frac{d^2}{4} + x^2 \right)^{3/2} = 2 \cdot \left(\frac{d^2}{4} + x^2 \right)^{3/2}$$

22

$$q = -300e$$

$$r = 0,04m$$



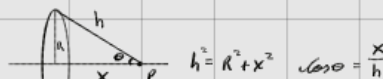
$$2\pi r = 360$$

$$x = 90$$

$$800 \cdot r = 260 \cdot x$$

$$x = \frac{800 \cdot r}{260} = 0,027$$

$$\lambda = \frac{q}{L} = \frac{-300 \cdot (1,6 \cdot 10^{-19})}{0,027} \approx 1,7 \cdot 10^{-15}$$

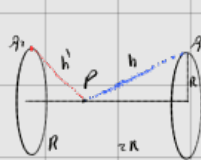


$$h^2 = R^2 + x^2 \quad \cos \theta = \frac{x}{h}$$

$$\int dE = \int \frac{k \cdot dq}{h^2} \cdot \cos \theta \rightarrow E = \frac{k \cdot \cos \theta}{h^2} \int dq \rightarrow E = \frac{k \cdot \frac{x}{\sqrt{R^2 + x^2}}}{R^2 + x^2} \cdot Q \rightarrow$$

$$E = \frac{k \cdot Q \cdot x}{(R^2 + x^2)^{3/2}}$$

23



$$dE = \frac{k \cdot dq}{h^2}$$

$$* \cos \theta = \frac{x}{h}$$

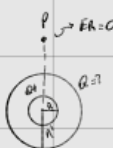
$$E_2 = E_1$$

$$h^2 = 4R^2 + R^2 = 5R^2$$

$$(h')^2 = R^2 + R^2 = 2R^2$$

$$\frac{k \cdot R \cdot Q_1}{(R^2 + h'^2)^{3/2}} = \frac{k \cdot 2R \cdot Q_2}{(R^2 + h^2)^{3/2}} \rightarrow \frac{Q_1}{Q_2} = \frac{2 \cdot (2R^2)^{3/2}}{(5R^2)^{3/2}} = 2 \cdot \left(\frac{2R^2}{5R^2} \right)^{3/2} = 2 \cdot \sqrt{\frac{2 \cdot 2}{5 \cdot 5}} = \frac{2 \cdot 2}{5} \cdot \sqrt{\frac{2}{5}} = \frac{4}{5} \cdot \sqrt{\frac{2}{5}}$$

28



$$E = \frac{k \cdot Q \cdot y}{(R^2 + y^2)^{3/2}}$$

$$R' = 3R$$

$$y = 2R$$

$$E_1 + E_2 = 0$$

$$\frac{k \cdot Q_1 \cdot 2R}{(R^2 + 4R^2)^{3/2}} + \frac{k \cdot Q_2 \cdot 2R}{(9R^2 + 4R^2)^{3/2}} = 0$$

$$\rightarrow \frac{k \cdot Q_1 \cdot 2R}{(5R^2)^{3/2}} = - \frac{k \cdot Q_2 \cdot 2R}{(13R^2)^{3/2}} \rightarrow Q_2 = - \frac{(13R^2)^{3/2}}{(5R^2)^{3/2}} Q_1 = - \left(\frac{13R^2}{5R^2} \right)^{3/2} Q_1 \approx -4,19 Q_1$$

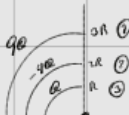
$$\therefore Q_2 = -4,19 Q_1$$

29

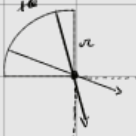
$$E = \frac{k \cdot Q \cdot y}{(R^2 + y^2)^{3/2}}$$

$$R = 10 \cdot a$$

$$Q = 2 \cdot 10^{-6} C$$



$$E_R = E_1 - E_2 + E_3$$



$$\lambda = \frac{Q}{s} \rightarrow dq = \lambda \cdot da$$

$$da = a \cdot d\theta, \text{ pois } s = a \cdot \theta$$

$$dE = \frac{k \cdot dq}{R^2} \rightarrow dE = \frac{k \cdot \lambda \cdot da}{R^2}$$

* Note que os campos não se anulam, apenas se anulam em x e y. Devemos integrar.

$$* \text{Após uma derivação o resultado} \rightarrow E_R = \sqrt{\left(\frac{k \cdot \lambda}{R^2}\right)^2 + \left(\frac{k \cdot \lambda}{R^2}\right)^2} = \frac{2k \cdot \lambda \cdot \sqrt{2}}{R^2}$$

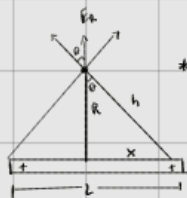
$$dE_x = \int \frac{k \cdot \lambda \cdot \cos \theta \cdot da}{R^2} = \frac{k \cdot \lambda}{R^2} \cdot [\sin \theta]_{\pi/2}^{\pi} = - \frac{k \cdot \lambda}{R^2}$$

$$dE_x = - \frac{k \cdot \lambda}{R^2}$$

$$dE_y = \int \frac{k \cdot \lambda \cdot \sin \theta \cdot da}{R^2} = \frac{k \cdot \lambda}{R^2} \cdot [-\cos \theta]_{\pi/2}^{\pi} = \frac{k \cdot \lambda}{R^2}$$

$$dE_y = \frac{k \cdot \lambda}{R^2}$$

32



$$R = 0,06 \text{ m} \quad L = 0,145 \text{ m}$$

$$Q = 7,81 \cdot 10^{-9} \text{ C}$$

* em q. x. lado.

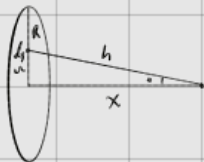
$$\cos \theta = \frac{R}{h}$$

$$\lambda = \frac{Q}{L} = \frac{7,81 \cdot 10^{-9}}{0,145} = 5,38 \cdot 10^{-8}$$

$$dq = \lambda \cdot dx = 5,38 \cdot 10^{-8} \cdot dx$$

$$dE = \frac{k dq}{h^2} \cos \theta \rightarrow dE = \frac{k \lambda dx R}{h^2} = \frac{k \lambda dx R}{h^2} \rightarrow E_R = k \lambda R \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{(R^2 + x^2)^{3/2}} dx$$

37



$$E_{\text{total}} = \frac{k \cdot Q \cdot x}{(R^2 + x^2)^{3/2}}$$

$$dq = \sigma \cdot dA = \sigma \cdot (2\pi r \cdot dr)$$

$$dE_{\text{total}} = \frac{k \cdot x \cdot \sigma \cdot 2\pi r \cdot dr}{(r^2 + x^2)^{3/2}} \rightarrow E_{\text{total}} = k \cdot x \cdot \sigma \cdot 2\pi \int_0^R \frac{r}{(r^2 + x^2)^{3/2}} dr \rightarrow E_{\text{total}} = k \cdot x \cdot \sigma \cdot 2\pi \cdot \left[\frac{1}{x} - \frac{1}{(R^2 + x^2)^{1/2}} \right]$$

$$A = \pi R^2 \rightarrow dA = 2\pi r \cdot dr \text{ (anulando)}$$

Compo elétrica:

$$\textcircled{a} \Delta V = -E \cdot \Delta x \rightarrow V_f - V_i = +20 \cdot 2 \rightarrow V_f = +30 \rightarrow \text{Sua } 30V$$

b) Sua em 3 m tem o maior campo elétrico u) $0 - 10 = -\text{Área do retângulo}$

$$\begin{aligned} -40 + 10 &= x \cdot (-20) \\ -30 &= x \cdot (-20) \quad \therefore x = 1,5 = 1,5 \\ x &= \frac{30}{20} = 1,5 \end{aligned}$$

$$\lambda = \frac{Q}{L}$$

$$V = k \cdot \lambda \cdot r_0$$

$$V = 9 \cdot 10^9 \cdot \frac{-25,6 \cdot 10^{-9}}{25,6 \cdot 10^{-2}} \approx -6,21V$$

29



$$dV = \frac{k dq}{r}$$

$$\sigma = \frac{Q}{A}$$

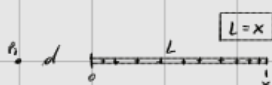
$$A = \frac{\pi \cdot 0,6^2}{4} = 0,28 \text{ m}^2$$

$$\sigma = Q$$

$$Q = 7,93 \text{ nC}$$

$$V = kQ \int_0^R \frac{1}{r} dr = kQ \left[\ln |r| \right]_0^R \rightarrow kQ \cdot \ln(R) - kQ \cdot \ln(0) =$$

30



$$dV = \frac{k dq}{x+d} \rightarrow dV = \frac{k \lambda dx}{x+d} \rightarrow V = k \lambda \int_0^L \frac{1}{x+d} dx \rightarrow V = k \lambda \cdot \ln \left| \frac{L+d}{d} \right|$$

$$\lambda = \frac{Q}{L} \rightarrow dq = \lambda dx$$

$$\int_0^L \frac{1}{x+d} dx = \int_d^{L+d} \frac{1}{u} du = \left[\ln |u| \right]_d^{L+d} = \ln |L+d| - \ln |d| = \ln \left| \frac{L+d}{d} \right|$$

$$u = L+d$$

$$du = dx$$

LEI DE GAUSS:

⑦ $E = 1800 \text{ N/C}$ $\theta = 35^\circ$

$$\Phi = E \cdot A \cdot \cos \theta = 100 \cdot (3,2 \cdot 10^{-2})^2 \cdot \cos 35^\circ = 1,5 \cdot 10^{-2} \text{ N} \cdot \text{m}^2/\text{C}$$

⑤ $q = 1,8 \cdot 10^6 \text{ C}$ $E = 0,05 \cdot 10^{-72}$

$$\phi = \frac{q}{\omega} = \frac{1,8 \cdot 10^{-6}}{8,85 \cdot 10^{-12}} = 0,203 \cdot 10^6 = 2,03 \cdot 10^5 \text{ N} \cdot \text{m} / \text{C}$$

④ Este superior: $\Phi = 39.9 \cdot \log 780 = -306 \text{ N.m}^2/\text{c}$

$$\Phi_{\text{total}} = -486 \text{ N.m}^2/\text{C}$$

Elab inferior: $\Phi: 20.9 \cdot \cos 180 = -180$

$$Q = -986.8, 89.70^{12} \approx 4307.70^{12}$$

⑦ $\lambda = 1,6 \cdot 10^{-79}$ $\phi = \frac{1,6 \cdot 10^{-79}}{8,85 \cdot 10^{-12}} = 0,18 \cdot 10^{-7} = 18 \cdot 10^{-9}$

(23)

$$\frac{E}{C_0} = E \cdot 20 \text{ A} \frac{E}{\lambda}$$

Relo superfície: $\frac{\Phi}{6} = \frac{18 \cdot 10^{-9}}{6} = 3 \cdot 10^{-9} \text{ N.m}^2/\text{C}$

$$E = 9.5 \cdot 10^9 \text{ N/C} \quad \phi = \frac{q_{\text{int}}}{\epsilon_0} = \oint E \cdot dA$$

$$\lambda = \frac{Q}{L} \rightarrow L = \frac{Q}{\lambda} \quad A = 2\pi R \cdot L = 2\pi R \cdot \frac{Q}{\lambda}$$

$$\lambda = E \cdot 2\pi \cdot n \cdot \epsilon_0$$

$$A = 4,5 \cdot 10^4 \cdot 2, \pi \cdot 2,885 \cdot 10^{-7} \text{ s}$$

$$\lambda = 5 \cdot 10^{-6}$$

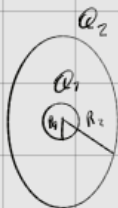
24) 

$$Q = \frac{q \cdot E}{r} = \frac{36 \cdot 10^{-17} \cdot 377 \cdot 10^6}{9,1 \cdot 10^{-31}} = 2,09 \cdot 10^{27}$$

$$\Phi = \frac{Q}{\epsilon_0} = \oint E \cdot dA$$

$$A = \frac{2\pi R \cdot Q}{6.90^{-6}} \rightarrow E = \frac{6.90^{-6}}{8.85 \cdot 10^{-12} \cdot 28.909} = 1,19.90^6$$

(24) $R_1 = 2,3 \text{ m}$ $R_2 = 93 \text{ m}$ $a = b$
 $\uparrow$
 $r_1 = 26 \text{ m}$ $r_2 = 6,5 \text{ m}$



$$Q_1 = 3, 4 \cdot 10^{-12}$$

$$Q_2 = -6, 8 \cdot 10^{-12}$$



$$\phi = \frac{q_{\text{int}}}{\epsilon_0} = \oint E \cdot dA$$

$$\frac{-3,4 \cdot 10^{-12}}{8,85 \cdot 10^{-12}} = E \cdot (2$$

$$d) |E| = 0,213 \text{ N/C}$$

b) O sentido da longo entro no texto.

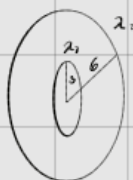
$$A = 2\pi r L \quad E = -0,273 \text{ N/C}$$



$$\frac{3,4 \cdot 10^{-12}}{8,85 \cdot 10^{12}} = E \cdot 2\pi \cdot 6,5 \cdot 10^{-3} \cdot 11 \quad \text{c) } E = 0,85 \text{ N/C}$$

d) Parafraze

(29)



* todos os pontos pedidos pelo quântico não dar sempre negativo

(36)

Eda plano: $2\pi\sigma = 0,25 \text{ N/C}$ $E_{\text{plano}} = E_{\text{disco}} = E_{\text{resultante}} \rightarrow E_R = 2\pi \cdot b \cdot \sigma \cdot \left(\frac{3}{\sqrt{3^2 + R^2}} \right) = 0,208 \text{ N/C}$

E do disco:

Diagrama de um disco de raio R e espessura h. Um elemento de área dA é considerado a uma distância x do ponto de observação. O ângulo entre a normal do elemento e a linha de observação é θ . A distância entre o elemento e o ponto de observação é r . A distância horizontal entre o elemento e o ponto de observação é x .

$$dE = \frac{k dq}{r^2} \cos \theta$$

$$dE = \frac{k \sigma \cdot 2\pi x \cdot dx \cdot \frac{x}{h}}{h^2} = \frac{k \sigma \cdot 2\pi x \cdot dx \cdot x}{h^3}$$

$$\sigma = \frac{dq}{dA} \rightarrow dq = \sigma \cdot dA$$

$$A = 2\pi x dx \text{ * área integral de } 0 \text{ a } R$$

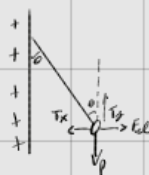
$$\cos \theta = \frac{x}{h} \quad h = \sqrt{x^2 + r^2}$$

$$E = \frac{k \sigma \cdot 2\pi x}{h^3} \int_0^R \frac{x}{(x^2 + r^2)^{3/2}} dx = k \sigma \cdot 2\pi x \cdot \left[-\frac{1}{\sqrt{x^2 + r^2}} \right]_0^R$$

$$= 2\pi \cdot b \cdot \sigma \cdot x \cdot \left[\left(-\frac{1}{x} \right) - \left(-\frac{1}{\sqrt{R^2 + x^2}} \right) \right] = 2\pi \cdot b \cdot \sigma \cdot \left(1 - \frac{x}{\sqrt{R^2 + x^2}} \right)$$

$$\therefore E_{\text{disco}} = 2\pi \cdot b \cdot \sigma \cdot \left(1 - \frac{x}{\sqrt{R^2 + x^2}} \right)$$

(47)



$$E = 2\pi \cdot b \cdot \sigma = 5,65 \cdot 10^{10}$$

$$T \cdot \cos \theta = m \cdot g$$

$$77547 \cdot \sin 30 = 2 \cdot 10^{-8} \cdot 5,65 \cdot 10^{10} \cdot \sigma$$

$$T = \frac{10 \cdot 10}{\cos 30} = 77547$$

$$\sigma = 5,7 \text{ C/m}^2$$

(46)

$$E = \frac{k q}{r^2} \rightarrow \frac{5,80 \cdot (0,03)^2}{9 \cdot 10^9} = a \rightarrow a = 2,2 \cdot 10^{-6}$$

* Gaussiano

(49)



$$Q_{\text{enc}} \rightarrow \oint \frac{Q}{a_{\text{enc}}} \rightarrow Q = \int \rho \cdot dV \rightarrow Q_{\text{enc}} = \int \left(\frac{A}{r^3} \right) \cdot \frac{4}{3} \pi (r^3) \cdot dr = 2\pi A (r^2 - a^2)$$

$$a_{\text{enc}} = \frac{4}{3} \pi r^3$$

$$\oint \frac{A}{r^3}$$

$$Q_{\text{enc}} = q + 2\pi A b^2 - 2\pi A \cdot a^2$$

$$E \cdot 4\pi b^2 = \frac{Q_{\text{enc}}}{\epsilon_0} \rightarrow \text{O preço todo o zero} \rightarrow E = \frac{k \cdot Q}{r^2} \rightarrow E = \frac{b \cdot (q + 2\pi A b^2 - 2\pi A \cdot a^2)}{r^2} \rightarrow \textcircled{III}$$

(II)

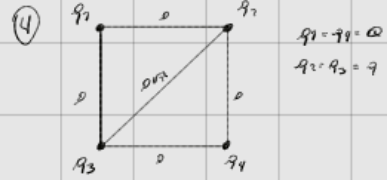
Se o campo for uniforme, então o cilindro $q - 2\pi A b^2 = 0$ com que ser repetido, com uma $A = 1,79 \cdot 10^{-11} \text{ C/m}^2$

SIMULADO GEMINI

$$(1) E = k \frac{q_1 q_2}{d^2} = 9 \cdot 10^9 \frac{26 \cdot 10^{-6} \cdot 47 \cdot 10^{-6}}{x^2} = 5,1 \quad x =$$

$$(3) U = k \frac{q_1 q_2}{r}$$

$$(2) d = 3,2 \cdot 10^{-3} \quad q = 9 \text{ nC/s} \quad m = 6,3 \cdot 10^{-9} \quad p) 6,3 \cdot 10^{-9} \cdot t = m \cdot q \rightarrow m_2 = 4,9 \cdot 10^{-7} \text{ kg}$$



$$(13) \text{radio} = \frac{r \cdot 3}{2} \quad \sigma = \frac{2,4 \cdot 10^{-6}}{\left(\frac{3 \cdot 3}{2}\right) \cdot 4 \cdot \pi} \approx 4,52 \cdot 10^{-7}$$

$$b) \text{ se forçaste, como campo potencial: } E = k \frac{q}{d^2} = 9 \cdot 10^9 \frac{2,4 \cdot 10^{-6}}{(965)^2}$$

$$(14) E = \frac{2k \cdot \lambda}{r} \rightarrow 4,5 \cdot 10^4 = \frac{2 \cdot \pi \cdot \lambda}{r} \rightarrow \lambda = \frac{4,5 \cdot 10^4}{k} \approx 5 \cdot 10^{-6}$$

$$(15) 4 \pi k \sigma = E$$

$$55 = 4 \pi \cdot 9 \cdot 10^9 \cdot \sigma$$

$$\sigma = \frac{55}{4 \pi \cdot 9 \cdot 10^9}$$